A-level

ENVIRONMENTAL SCIENCE

Paper 1 - Generated practice paper Set 31

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

• Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
• The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
• **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
• Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
• For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
• Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	**1 mark each** 1 mark for each of the 4 points below. **1 Gasification:** coal (including deposits too deep to mine) is burnt underground under controlled conditions to produce fuel gases (hydrogen, carbon monoxide, methane) **1** Allows deep coal to be exploited and gas can be piped / used in gas turbines or for hydrogen with CCS **1 Liquefaction:** coal is converted to liquid hydrocarbons directly with solvents or via gasification and chemical conversion **1** Provides liquid fuels for vehicles / aircraft that solid coal cannot supply **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **CCS** → capturing carbon dioxide from flue gases and storing it underground	4
01.2	**1 mark each** 1 mark for each of the 2 points below. **1** A solid, ice-like crystalline structure in which methane is trapped within water ice **1** Found at low temperatures (polar regions / permafrost) or high pressure in ocean sediments around continental margins **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2

Do not write outside the box

Question	Marking guidance	Marks
01.3	**1 mark each** 1 mark for each of the 4 points below. **1 Water heating:** hot water pumped into sediments melts the hydrate releasing methane **1 Depressurisation:** drilling reduces pressure so methane dissociates from the crystals **1 CO2 injection:** CO2 bonds more strongly to the ice and displaces methane (could double as CCS) **1 Risks:** methane not collected escapes to the atmosphere - powerful greenhouse gas / positive feedback – seabed ecology poorly understood – sediment instability / landslides **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **CCS** → capturing carbon dioxide from flue gases and storing it underground • **positive feedback** → a change that makes itself larger / a self-reinforcing cycle	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 187, 190; AQA spec 3.3.4.1.	
02.1	**Calculation** 1 mark for the correct answer; working does not have to be shown. **1** (120 + 95) - (2 + 45) = 168 t ha^{-1} **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (t ha^{-1}) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	1
02.2	**1 mark each** 1 mark for each of the 2 points below. **1** Annual ploughing aerates the soil so aerobic decomposers break down dead organic matter faster, releasing CO2 **1** Harvest removes biomass so less dead organic matter is returned – pasture roots and litter constantly add organic matter **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2

Do not write outside the box

Question	Marking guidance	Marks
02.3	**Any 2 of 4** 1 mark each for any 2 of the 4 points below, in any order, maximum 2. **1** Low / zero tillage **1** Adding manure, compost or mulch **1** Growing cover crops / leaving crop residues **1** Converting to long-term grass or tree crops in rotation **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 148-151; AQA spec 3.2.4.2.	
03.1	**1 mark each** 1 mark for each of the 2 points below. **1** Deaths rise as smoke concentration rises, peaking at about 930 per day on 8 Dec when smoke was highest (4.5 mg m^{-3}) **1** Deaths fall as smoke falls but remain above the starting level for several days (time lag / delayed effect) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
03.2	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** (930 - 250) / 250 x 100 **1** = 272% **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (%) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	2

Do not write outside the box

Question	Marking guidance	Marks
03.3	**1 mark each** 1 mark for each of the 3 points below. **1 Anticyclonic conditions:** clear skies allowed rapid radiation of heat from the ground at night so the ground and air near it cooled **1** Low wind velocity so air layers did not mix – London lies in a valley / basin where cold dense air collected: a temperature inversion formed **1** Smoke from coal fires was trapped below the inversion – moist air cooled to its dew point forming fog – smoke + fog = smog – the smog's high albedo prolonged the inversion **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **temperature inversion** → a layer of warm air above cooler air that traps pollutants • **albedo** → reflectivity / the proportion of solar radiation reflected / how much sunlight is bounced back **Do not credit** • 'ice reflects the heat back' - albedo is the reflection of incoming short-wave solar radiation, before it is absorbed	3
03.4	**1 mark each** 1 mark for each of the 2 points below. **1** Smoke particles (PM10) are inhaled deeply and irritate the lungs / kill cilia so particles and bacteria are not cleared, causing bronchitis / infections **1** These groups have weaker lungs / immune systems / less reserve so the extra stress (with sulfur dioxide acting synergistically) was fatal **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **PM10** → fine solid particles / smoke and dust	2
03.5	1 mark for the point below. **1** Clean Air Act (1956) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 245-247; AQA spec 3.4.3.2.1.	
04.1	**1 mark each** 1 mark for each of the 3 points below. **1** Spoil contains residual sulfide ores (eg iron pyrites, zinc sulfide) **1** These react with oxygen in the air spaces between the waste particles (oxidation) **1** Producing metal oxides and oxides of sulfur which dissolve in drainage water to form sulfuric acid **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3

Do not write outside the box

Question	Marking guidance	Marks
04.2	**1 mark each** 1 mark for each of the 3 points below. **1** The low pH (3.4) directly damages exposed cells / denatures proteins and dissolves calcium-based exoskeletons **1** Low pH increases the solubility of zinc (0.02 to 6.5 mg dm^{-3}) which is toxic (inhibits enzymes) **1** Only a few tolerant taxa survive – the sensitive mayfly larvae (indicator of clean water) are eliminated – dissolved oxygen is not the limiting factor here **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **limiting factor** → the factor in shortest supply that holds the rate or population back	3
04.3	**1 mark each** 1 mark for each of the 2 points below. **1** Dilution by tributaries / more water **1** Neutralisation by alkaline rocks (limestone) in the stream bed – zinc precipitates / adsorbs onto sediments as pH rises **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first 2 - the question asks for 2, so a list of extras cannot pick up the marks.	2
04.4	**1 mark each** 1 mark for each of the 2 points below. **1** Collect the drainage water and pass it through crushed limestone / add lime to neutralise the acid **1** Higher pH also reduces the solubility of toxic metals so they precipitate and can be removed **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 273-275; AQA spec 3.4.3.2.9-3.4.3.2.10.	
05.1	**1 mark each** 1 mark for each of the 3 points below. **1** A turbofan at the front forces bypass air around the inner cowling / core jet **1** The bypass air smooths the flow of the turbulent hot exhaust gases, reducing the turbulence that creates noise **1** The higher the ratio of bypass air to jet exhaust the quieter the engine (Trent 1000 quietest, Olympus noisiest) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3

Do not write outside the box

Question	Marking guidance	Marks
05.2	**Any 2 of 6** 1 mark each for any 2 of the 6 points below, in any order, maximum 2. **1** Chevron (serrated) nozzles mix bypass air with surrounding air more smoothly **1** Acoustic liners in the engine cowlings absorb noise **1** Hush kits fitted to older low-bypass engines **1** Fairings on undercarriage and wing-flap hinges reduce turbulence – welded rather than riveted panels **1** Lighter composite aircraft need less thrust **1** Blended wing with engines on top so the body shields the ground **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2
05.3	**1 mark each** 1 mark for each of the 4 points below. **1** Constant (3°) descent avoids periods of high thrust in a stepped descent and keeps aircraft higher for longer **1** Night flights (especially take-offs, which are noisier) are restricted / banned when people are sleeping and background noise is low **1** Noisier aircraft pay higher charges / use up more quota points and fines for exceeding limits, so airlines are encouraged to buy quieter aircraft **1** More time to slow down after landing so wheel brakes can be used instead of noisy reverse thrusters **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **quota** → a legal limit on the mass or number that may be caught **Do not credit** • quotas described as solving overfishing on their own - undersized fish are still caught and discarded	4
05.4	1 mark for the point below. **1** Engines run at higher power / maximum thrust and the steep climb angle needs full power (although the aircraft climbs quickly out of earshot) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 288-292; AQA spec 3.4.3.2.13.	

Do not write outside the box

Question	Marking guidance	Marks
06.1	**Calculation** 1 mark for the correct answer; working does not have to be shown. **1** (44 - 8) / 50 = 0.72 m yr^{-1} **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (m yr^{-1}) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	1
06.2	**1 mark each** 1 mark for each of the 2 points below. **1** Abstraction (for irrigation) exceeds the rate of natural recharge by infiltration / percolation of precipitation **1** The dynamic equilibrium between inflow and outflow has been upset so the stored volume declines **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **Abstraction** → water taken out of a river, lake or aquifer • **infiltration** → water soaking into the soil surface • **percolation** → water moving downwards through soil or rock	2
06.3	**1 mark each** 1 mark for each of the 3 points below. **1 Porosity:** a high proportion of the rock volume is space that can hold water (chalk, limestone, sandstone) **1 Permeability:** the spaces are interconnected and large enough for water to flow through (clay is porous but not permeable) **1** Impermeable rock (granite, clay) beneath to stop water escaping, and permeable rock / soil above to allow recharge **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **aquifer** → a permeable rock that stores groundwater / underground water store • **Permeability** → how easily water passes through / lets water through • **Impermeable** → does not let water through	3

Do not write outside the box

Question	Marking guidance	Marks
06.4	**Any 4 of 5** 1 mark each for any 4 of the 5 points below, in any order, maximum 4. **1 Salt water incursion:** seawater flows into the aquifer to replace the fresh water, making it unusable for irrigation (osmotic dehydration of crops) **1 Subsidence:** emptied pore spaces no longer support the rock so it compacts, damaging buildings and pipes (eg Mexico City) **1** Drying up of groundwater-fed rivers, lakes and wetlands as the water table falls below their beds **1 Ecological impacts:** plants needing high water availability die – wetland animals lost – food chain effects **1 Reduced supplies:** abstraction may suddenly have to stop after years of hidden decline **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **aquifer** → a permeable rock that stores groundwater / underground water store • **abstraction** → water taken out of a river, lake or aquifer **Do not credit** • Points after the first four - the question asks for four, so a list of extras cannot pick up the marks. • 'energy is lost between trophic levels' on its own - say how it is lost: respiration, excretion, egestion and uneaten parts	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 119-121; AQA spec 3.2.2.1-3.2.2.2.	
07.1	**1 mark each** 1 mark for each of the 2 points below. **1** About 8 nights **1** The running mean fluctuates widely with few samples but stabilises at 21-22 from 8 nights onwards – extra samples do not change it significantly **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **reliable** → repeatable / gives consistent results when repeated **Do not credit** • 'repeat the readings to make them more accurate' - repeats improve reliability; accuracy depends on the method and the instrument	2
07.2	**1 mark each** 1 mark for each of the 3 points below. **1** Moth activity varies with temperature, wind, rain and season (adults only fly for part of the year) so catches vary independently of population size **1** Sampling at different conditions / times would make results not comparable **1** Timing between samples should be chosen to detect the temporal variation of interest (eg weekly through the season) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3

Do not write outside the box

Question	Marking guidance	Marks
07.3	**1 mark each** 1 mark for each of the 2 points below. **1 Accuracy:** how close a recorded result is to the real value **1 Precision:** the smallest interval between possible recorded values (eg a ruler marked in mm is more precise than one in cm) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **accuracy** → close to the true value • **precision** → readings close together / fine resolution	2
07.4	**1 mark each** 1 mark for each of the 3 points below. **1** A null hypothesis (eg no difference / no correlation) can be tested statistically and accepted or rejected at a stated confidence level **1** Standardised techniques ensure results collected at different times, places or by different people are comparable **1** Other variables are controlled or monitored so their effect can be assessed – allows the study to be repeated / checked (reliability) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **null hypothesis** → a statement that there is no significant difference or correlation • **correlation** → a relationship in which the two variables change together • **reliability** → repeatable / gives consistent results when repeated **Do not credit** • a correlation presented as proof of cause - say that it suggests a link that further work would have to test • 'repeat the readings to make them more accurate' - repeats improve reliability; accuracy depends on the method and the instrument	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 390-393; AQA spec 3.7.1.	
08.1	**1 mark each** 1 mark for each of the 2 points below. **1** Minerals can only be exploited where they were deposited (no choice of location), unlike housing, roads or ports **1** Land take is larger than the mine void: buildings, access roads, overburden and spoil heaps, buffer zones – open-cast mines take far more land than deep mines **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2

Do not write outside the box

Question	Marking guidance	Marks
08.2	**1 mark each** 1 mark for each of the 5 points below. **1 Dust:** water sprays on roads / blasting areas make particles heavier and clump so they settle **1 Noise:** baffle mounds / embankments absorb and deflect – blasting at set, predictable daytime hours – quieter machinery **1 Turbid water:** sedimentation lagoons where water stands until solids settle before discharge **1 Leachate:** collect drainage water and pass through crushed limestone to neutralise acid and immobilise toxic metals **1 Amenity:** landscaping and tree planting to screen the mine – restoration plans agreed as a condition of planning permission **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **Leachate** → polluted liquid draining out of waste	5
08.3	**1 mark each** 1 mark for each of the 3 points below. **1** Landscaping to reduce gradients, compaction, drainage to prevent waterlogging, adding soil / nutrients and planting vegetation whose roots bind the surface **1** At Aberfan a waterlogged coal-mine spoil heap became unstable and slid into the village **1** Killing 144 people, mainly children in the primary school **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **waterlogging** → soil pores filled with water so there is no air	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 139-142; AQA spec 3.2.3.6.	
09.1	**1 mark each** 1 mark for each of the 3 points below. **1** Radiation absorbed by tissue produces free radicals (ions with unpaired electrons), mostly from water, which are highly reactive **1** Free radicals damage DNA / chromosomes in the nucleus causing mutations (cancer, inherited effects) - rapidly dividing cells (skin, gut, bone marrow) most affected **1** Damage to other cell parts can stop the cell functioning or kill it – low damage may be repaired by the intact nucleus **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **ionising radiation** → radiation with enough energy to remove electrons and form ions	3

Do not write outside the box

Question	Marking guidance	Marks
09.2	**1 mark each** 1 mark for each of the 2 points below. **1 Acute:** effects appearing quickly after a large dose over a short time (radiation sickness, damage to bone marrow / immune system / gut, haemorrhage, rapid death) **1 Chronic:** effects appearing slowly, in proportion to the dose received over a long period (cancers, cataracts) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
09.3	**1 mark each** 1 mark for each of the 2 points below. **1 Exposure:** absorption of ionising radiation from a source the person is close enough to **1 Contamination:** physically carrying radioactive material on or in the body, so exposure continues wherever the person goes – sealed sources can expose but not contaminate **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **ionising radiation** → radiation with enough energy to remove electrons and form ions	2
09.4	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** Sieverts = grays x radiation weighting factor (20 for alpha) **1** 0.002 x 20 = 0.04 Sv (40 mSv) **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (Sv) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	2
09.5	1 mark for the point below. **1** The activity of a source (number of decays per second) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **becquerel** → one radioactive decay per second	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 301, 303; AQA spec 3.4.3.2.14.	

Do not write outside the box

Question	Marking guidance	Marks
10.1	**Levels of response** Decide the level from the descriptors first, then use the indicative content to place the mark within that level. The points below are not a checklist. **Level 3 (7-9 marks)**: Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. **Level 2 (4-6 marks)**: Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. **Level 1 (1-3 marks)**: Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. **0 marks**: No relevant content. **Indicative content - a guide to placing the mark, not a checklist** • Neonicotinoids are neurotoxins similar to nicotine – they inhibit the neurotransmitter acetylcholine • High toxicity to insects but low toxicity to vertebrates because they cannot cross the vertebrate blood-brain barrier • **Very toxic to bees:** sub-lethal doses reduce navigation ability, foraging and immunity to disease – linked to colony decline • **Water soluble and moderately persistent:** remain in soil, absorbed by roots and translocated (systemic), so present in pollen and nectar • Persistence may allow build-up in aquifers / soil – broken down fastest in sunlight and by soil microbes • Synergism with some fungicides increases toxicity to bees • Exposure greatest when sprayed onto crops or when seed-coating dust is blown by wind • **Control:** seed coating rather than spraying – spraying when crops are not in flower / at night – still days to reduce drift • Restrictions on use (EU bans on outdoor use on flowering crops) – use of alternative pest control (cultural, biological, integrated control) • **Evaluation:** reduced pesticide use but neonicotinoid alternatives (pyrethroids) also harm pollinators – trade-off with crop yields **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • Any accurate, relevant material that is not in the indicative content. • **persistent** → does not break down / stays in the environment for a long time • **Synergism** → two pollutants together doing more damage than the sum of each alone **Do not credit**	9
	• pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on	
	Model answer - Level 3 (8 marks)	

Do not write outside the box

Question	Marking guidance	Marks
	Neonicotinoids are neurotoxins related to nicotine that block the neurotransmitter acetylcholine. They are highly toxic to insects but of low toxicity to vertebrates because they cannot cross the vertebrate blood-brain barrier - the specificity that made them popular. Bees, however, are insects: even sub-lethal doses impair navigation, foraging and immunity, and this has been linked to colony decline. Because the compounds are water-soluble and moderately persistent they remain in the soil, are absorbed by roots and translocated through the plant (systemic action), so they appear in pollen and nectar that bees collect; persistence may also allow build-up in soils and aquifers, although sunlight and soil microbes break them down. Some fungicides act synergistically with them, increasing toxicity to bees.	
	Exposure is greatest when crops are sprayed or when dust from coated seeds is blown by wind, so impacts are reduced by using seed coatings instead of sprays, spraying only when crops are not in flower or at night when bees are not foraging, and applying on still days to limit drift. The EU has banned outdoor use on flowering crops, and integrated control with cultural and biological methods reduces the need for insecticides altogether.	
	These measures reduce bee exposure, but the alternatives farmers turn to, such as pyrethroids, also harm pollinators, and restrictions can reduce yields, so the trade-off between crop protection and pollination services remains.	
	Weaker answer - Level 2 (4 marks)	
	Neonicotinoids are insecticides that kill insects by affecting their nervous system. They are systemic so they get into the whole plant including the pollen, and bees pick them up when they collect pollen and nectar. This harms bees and can cause colonies to die. To reduce the impact, farmers can spray when crops are not flowering, and some neonicotinoids have been banned in the EU. Other methods of pest control could be used instead.	
	What makes the difference	
	• Level 2 links the systemic property to bee exposure correctly but omits the other properties (solubility, persistence, specificity via the blood-brain barrier, synergism) that the question's 'properties' demands.	
	• Level 3 explains the sub-lethal effects (navigation, foraging, immunity) rather than 'harms bees'.	
	• Level 3 includes the trade-off evaluation (alternatives also harm pollinators, yield costs) that the mark scheme rewards.	

Do not write outside the box

Question	Marking guidance	Marks
10.2	**Any 1 of 6** 1 mark each for any 1 of the 6 points below, in any order, maximum 1. **1** Biological control (introduced predators / pathogens) **1** Predator habitats (hedgerows, beetle banks) **1** Crop rotation **1** Companion / barrier crops **1** Sterile male technique / pheromone traps **1** Genetic resistance **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **Biological control** → using a natural predator, parasite or pathogen of the pest **Do not credit** • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 265-266; AQA spec 3.4.3.2.8.	

Do not write outside the box

Question	Marking guidance	Marks
11.1	**To what extent can renewable energy resources replace fossil fuels in meeting the energy needs of a modern society?** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • Fossil fuel features that shaped society: energy density, storage, transport, high temperatures, liquid vehicle fuels, established infrastructure • **Renewable features:** intermittent / unpredictable (wind, solar) vs predictable (tidal) vs controllable (HEP, biofuel, geothermal) – low energy density – location – low pollution but manufacturing impacts • Technologies and developments for each resource with examples – new technologies improving usability • **Storage and grid management:** pumped storage, batteries, hydrogen economy, P2G, V2G, thermal storage, peak shaving – the role of electricity as a secondary fuel • **Economics and politics:** development costs, interest, subsidies, external costs of fossil fuels, falling renewable costs • Energy conservation as part of the solution – sustainability principles - natural systems run on low-density renewable energy • **Judgement:** electricity largely yes	25
	– heat, aviation, shipping and high-temperature industry harder – transition period with nuclear / CCS *Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.* **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 175-235; AQA spec 3.3.2-3.3.4.	

Do not write outside the box

Question	Marking guidance	Marks
11.2	**Discuss the extent to which the study of ozone depletion illustrates how scientific investigation can lead to the successful solution of an environmental problem.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • **Stages of investigation:** identifying the issue (Halley Station UV data, Dobson unit measurements), hypothesis formation (Rowland-Molina 1974: persistence, UV dissociation, chlorine reactions), data collection (ground, satellite, balloon and aircraft air sampling detecting ClO), analysis of spatial, temporal and altitude variability, proposal of solutions, enactment • **Why Antarctica:** low stratospheric temperatures, ice crystals, stratospheric clouds, polar vortex, spring sunlight • **Effects of UVB:** skin cancer, cataracts, DNA damage, plant and marine damage - motivating action • **Solutions:** Vienna Convention, Montreal Protocol - phase-out and bans, alternative processes (pump sprays), alternative materials (HCFCs, HFCs, hydrocarbons, HFAs), collection and incineration of CFCs, fund for developing countries • **Evaluation of success:** ozone hole area / depth stabilising, CFC levels falling	25
	– limitations - persistence, remaining ODSs, HFCs as greenhouse gases, decades to full recovery • Comparison with other atmospheric problems (climate change, acid rain) - why ozone was easier – role of peer review, international cooperation, availability of substitutes • Conclusion on the extent to which it is a model for other issues *Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.* **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 108-113; AQA spec 3.2.1.3.	

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